

Mock Exam: Data Management & Business Intelligence

90 multiple choice questions. Choose one answer (A-D) per question.

Business Intelligence

1. What does BI stand for? A. Business Information B. Business Intelligence C. Binary Index D. Budget Integration
2. What is the main purpose of BI? A. Replace databases B. Support decision making with data C. Store raw logs only D. Increase file system speed
3. BI is not an end in itself. It should be: A. Isolated from business processes B. Fully integrated into decision-making processes C. Used only for compliance D. Limited to IT departments
4. A data-driven culture means: A. Decisions rely on data and analytics B. Decisions rely on intuition only C. Data stored in spreadsheets only D. No data sharing allowed
5. Data silos are: A. Shared company-wide data lakes B. Isolated data systems across departments C. Backups stored offsite D. Encrypted databases
6. Shadow BI refers to: A. Official BI tools used by IT B. Unofficial BI practices outside central governance C. Historical BI reports D. BI training courses
7. Descriptive analytics answers: A. What should we do? B. Why did it happen? C. What happened? D. What will happen?
8. Operational BI (OBI) focuses on: A. Long-term research only B. Real-time or near-real-time operational decisions C. Replacing data warehouses D. Writing code without data
9. Business-data disconnect means: A. Data is encrypted B. Data insights are misaligned with business goals C. Too many dashboards D. Only IT can access data
10. Diagnostic analytics answers: A. What happened? B. Why did it happen? C. What will happen? D. What should we do?
11. The decision-action gap means: A. Too much data B. Decisions are made without data C. Insights exist but are not acted on D. Data is deleted too early
12. Predictive analytics answers: A. What happened? B. Why did it happen? C. What will happen? D. What should we do?
13. Organizational silos mainly refer to: A. Too many dashboards B. Lack of communication between departments C. Too many backups D. Slow internet
14. Prescriptive analytics answers: A. What happened? B. Why did it happen? C. What will happen? D. What should we do?
15. Netflix collects viewing data mainly to: A. Reduce storage costs only B. Understand engagement and guide content decisions C. Replace all BI tools D. Avoid dashboards

Python

1. Which is a valid Python variable name? A. total_sales B. 2nd_value C. class D. total-sales
2. What is the result of `len([1, 2, 3, 4])`? A. 3 B. 4 C. 5 D. Error
3. Which type stores key-value pairs? A. list B. tuple C. dict D. set
4. What does this print? `for i in range(3): print(i)` A. 1 2 3 B. 0 1 2 C. 0 1 2 3 D. 3 2 1
5. What is the result of `"data".upper()`? A. "Data" B. "DATA" C. "data" D. Error
6. What does `my_list.append(5)` do? A. Adds 5 to the start B. Adds 5 to the end C. Removes 5 D. Sorts the list

7. What prints? `x = 5 if x > 3: print("yes") else: print("no")` A. yes B. no C. 5 D. Error
8. What does this return? `def add(a, b): return a + b add(2, 3)` A. 2 B. 5 C. 6 D. Error
9. For `items = [10, 20, 30]`, what is `items[0]`? A. 10 B. 20 C. 30 D. Error
10. What does `import math` do? A. Deletes a module B. Imports the math module for use C. Creates a new file D. Runs a script
11. What is `sum([1, 2, 3])`? A. 3 B. 5 C. 6 D. 7
12. What is `{"a": 1, "b": 2}["b"]`? A. 1 B. 2 C. "b" D. Error
13. What is `[x * 2 for x in [1, 2, 3]]`? A. [2, 4, 6] B. [1, 2, 3] C. [2, 3, 4] D. Error
14. What is `"a,b,c".split(",")`? A. ["a,b,c"] B. ["a", "b", "c", ""] C. ["a", "b", "c"] D. Error
15. For `items = ["a", "b", "c"]`, what is `items[-1]`? A. "a" B. "b" C. "c" D. Error

Database Architectures

1. What is a task of a file system? A. Bring order to files/folders and metadata B. Train ML models C. Replace SQL D. Encrypt all data
2. CRUD stands for: A. Create, Read, Update, Delete B. Copy, Run, Undo, Deploy C. Cache, Restore, Update, Drop D. Create, Replace, Use, Dump
3. The no free lunch theorem implies: A. No backups are required B. Performance tradeoffs exist (faster reads can mean slower writes) C. Data is always consistent D. Storage is free
4. In a row-oriented database, accessing by row index is typically: A. Fast (direct access) B. Slow because it scans all rows C. Impossible D. Only possible in the cloud
5. Searching by column value in a row-oriented DB is slow because: A. Data is encrypted B. You often scan rows to compare values C. Rows are sorted by that column by default D. Indexes are free and automatic
6. A transaction is: A. A single query only B. A unit of work treated as one C. A backup file D. A schema definition
7. An index is mainly used to: A. Slow down reads B. Speed up searches and lookups C. Replace primary keys D. Store images
8. Binary search complexity is: A. $O(1)$ B. $O(n)$ C. $O(\log n)$ D. $O(n^2)$
9. A tradeoff of using indexes is: A. No storage needed B. Slower reads only C. Extra storage and slower writes or updates D. Data becomes raw
10. ACID stands for: A. Atomicity, Consistency, Isolation, Durability B. Accuracy, Control, Index, Data C. Access, Compute, Integrate, Delete D. Archive, Copy, Inspect, Deploy
11. A primary key: A. Allows duplicate values B. Uniquely identifies each row C. Stores blobs D. Is optional for tables
12. A foreign key: A. Uniquely identifies rows in the same table B. References a primary key in another table C. Stores passwords D. Replaces indexes
13. Document databases typically store: A. Fixed rows and columns only B. JSON-like documents with flexible schema C. Only CSV files D. Only images
14. SQL JOIN is used to: A. Back up data B. Combine rows from multiple tables C. Delete a table D. Change column types
15. Many-to-many relationships usually use: A. A linking table B. A single column only C. A file folder D. A view only

Data Management

1. A data warehouse is mainly: A. Raw data storage without transformations B. Central hub for cleaned structured data loaded via ETL C. Only used for image storage D. A file system replacement
2. ETL stands for: A. Extract, Transform, Load B. Encode, Transfer, List C. Extract, Translate, Link D. Export, Test, Log

3. A data lake typically: A. Stores only structured data B. Loads data without transformations C. Requires fixed schema D. Rejects text or audio
4. Staging tables are: A. Final dashboard tables B. Cleaned reusable tables for multiple marts C. Only for backups D. Raw unprocessed logs
5. A common risk of data lakes is: A. Data becoming a data swamp due to lack of structure B. Too much normalization C. Only supporting SQL D. Overly strict schemas
6. A mart table is: A. A raw data dump B. A dashboard-specific curated table C. A file system index D. A log archive
7. A data lake advantage is: A. Low flexibility B. Supports many data types and formats C. Requires heavy transformations D. Only for small datasets
8. A hybrid data lake: A. Is a data warehouse only B. Combines data lake and data warehouse but adds complexity C. Removes the need for storage D. Works only on-prem
9. On-premises vs cloud: which is correct? A. On-prem uses provider data centers only B. Cloud means local servers only C. On-prem uses company-owned infrastructure; cloud uses hosted services D. They are identical
10. A lakehouse: A. Combines data warehouse and data lake with lower complexity B. Is identical to a file system C. Avoids query engines D. Stores only logs
11. A good ETL process includes: A. No documentation B. Monitoring, testing, and lineage/traceability C. Only manual steps D. Deleting errors
12. Which is a common cloud service model? A. KaaS B. IaaS, PaaS, or SaaS C. DaaS only D. FaaS only
13. A high cost driver in a pure data warehouse can be: A. Schema changes requiring large backfills B. No storage C. No transformations D. Only text data
14. A data lake often needs: A. No query tooling B. Extra tooling for query and processing C. Only spreadsheets D. Only SQL views
15. A potential disadvantage of a lakehouse is: A. It cannot store structured data B. A standalone query engine may be needed C. It blocks ETL D. It only works on-prem

Introduction to AI

1. Artificial Intelligence is defined as: A. Simulation of intelligence, especially learning B. Only robots C. Only databases D. Only hardware
2. Era 1 (pre-deep learning) focused on: A. Rules, logic, expert systems B. Large-scale LLMs only C. Purely random search D. No data
3. Regression outputs: A. A category label B. A number C. A document D. An image
4. Supervised learning uses: A. No labels B. Labels for training C. Only rules D. Only random actions
5. Mean squared error (MSE) is: A. A type of database B. A common loss for regression C. A clustering method D. A storage format
6. Unsupervised learning example is: A. Demand forecasting B. Customer segmentation C. Spam classification D. Price prediction
7. Classification outputs: A. A category label B. A number C. A file D. A graph
8. Reinforcement learning learns by: A. Labels only B. Rewards and feedback C. Fixed rules only D. SQL queries
9. Gradient descent updates parameters by: A. Moving in the direction of higher loss B. Moving opposite the gradient C. Random guessing only D. No updates
10. k-means steps are: A. Initialize, assign, update, repeat B. Encode, train, deploy, monitor C. Select, delete, migrate, backup D. Sort, join, filter, group
11. k-means output is: A. A single number B. Cluster assignment for each point C. A decision tree D. A SQL query
12. Supervised learning example: A. Customer churn prediction B. Discovering clusters C. Anomaly grouping without labels D. Random guessing
13. Unsupervised learning uses: A. Inputs with labels y B. Only rewards C. Inputs without labels D. Only rules

14. The learning rate in gradient descent is: A. A type of dataset B. The step size for updates C. A measure of accuracy D. A storage limit
15. Training typically repeats until: A. Loss stops improving B. The dataset is deleted C. Labels are removed D. Queries fail

Generative AI and Agents

1. Generative AI primarily: A. Creates synthetic data or content B. Only stores data C. Only classifies text D. Only deletes data
2. An LLM mainly predicts: A. Database schema B. The next token C. A SQL join D. A file path
3. The context window is: A. The model's training dataset size B. The amount of text the model can use at once C. The output length limit only D. The GPU memory size
4. A hallucination is: A. Perfect memory B. Confident but incorrect output C. A database index D. A training label
5. A GAN has: A. Encoder and decoder only B. Generator and discriminator C. Router and cache D. Tokenizer only
6. Transformers use attention to: A. Ignore all context B. Capture dependencies and focus on relevant tokens C. Remove tokens D. Encrypt data
7. Encoder-decoder models are often used to: A. Map input sequences to output sequences (e.g., translation) B. Store images only C. Run SQL D. Compress files only
8. Vector databases exist mainly for: A. Keyword search only B. Semantic similarity search in embeddings C. Storing backups D. Running ETL
9. An embedding is: A. A high-dimensional vector representation of data B. A primary key C. A file extension D. A SQL query
10. RAG stands for: A. Retrieve, Augment, Generate B. Read, Analyze, Graph C. Random, Approximate, Guess D. Reduce, Aggregate, Group
11. A ReAct agent combines: A. Reasoning and acting with observations B. Only training C. Only storage D. Only encryption
12. A simple agent loop includes: A. Goal, plan, tools, feedback loop B. Tables, rows, columns, keys C. Images, audio, video only D. Compile, link, run, exit
13. Tool calls are: A. External function interfaces the model can invoke B. Only UI buttons C. SQL backups D. Static images
14. MCP stands for: A. Model Context Protocol B. Machine Code Parser C. Multi Cloud Platform D. Model Cache Process
15. ACP stands for: A. Agent Communication Protocol B. Automatic Code Pipeline C. Access Control Policy D. Agent Cloud Platform

Answer Key 1-B, 2-B, 3-B, 4-A, 5-B, 6-B, 7-C, 8-B, 9-B, 10-B, 11-C, 12-C, 13-B, 14-D, 15-B, 16-A, 17-B, 18-C, 19-B, 20-B, 21-B, 22-A, 23-B, 24-A, 25-B, 26-C, 27-B, 28-A, 29-C, 30-C, 31-A, 32-A, 33-B, 34-A, 35-B, 36-B, 37-B, 38-C, 39-C, 40-A, 41-B, 42-B, 43-B, 44-B, 45-A, 46-B, 47-A, 48-B, 49-B, 50-A, 51-B, 52-B, 53-B, 54-C, 55-A, 56-B, 57-B, 58-A, 59-B, 60-B, 61-A, 62-A, 63-B, 64-B, 65-B, 66-B, 67-A, 68-B, 69-B, 70-A, 71-B, 72-A, 73-C, 74-B, 75-A, 76-A, 77-B, 78-B, 79-B, 80-B, 81-B, 82-A, 83-B, 84-A, 85-A, 86-A, 87-A, 88-A, 89-A, 90-A